W-S-90C
June 14, 1995
SUPERSEDING
W-S-0090B (GSA-FSS)
October 23, 1967

FEDERAL SPECIFICATION

SAW, RECIPROCATING, PORTABLE, ELECTRIC

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric reciprocating saws used for sawing sheet metal, mild steel, aluminum, brass, wood, and other materials.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The portable electric reciprocating saws covered by this specification shall be of the following types and classes, as specified, (see 6.1):

Type I - Horizontal Drive (In-Line)
Class 1 - One Speed, General Purpose
Class 2 - Multi-Speed, Metal and Woodcutting

Type II - Vertical Drive, General Purpose

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

FED-STD-102	-	Preservation, Packaging and Packing Levels
FED-STD-123	-	Marking for Domestic Shipment (Civilian Agencies)
FED-STD-H28	-	Screw-Threads Standards for Federal Services

Federal Specification

GGG-B-425	-	Blade, Reciprocating, Saw
PPP-P-40	-	Packaging and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

FSC 5130

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Military Specifications:

MIL-P-116 - Preservation, Methods of

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P. O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

Underwriters' Laboratories, Inc. (UL)

UL 45 - Electric Tools

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

3. REQUIREMENTS

3.1 Conformance to Underwriters' Laboratories (UL), Inc. standards. The saw shall comply with the requirements of the specified Underwriters' Laboratories (UL) standards. The contractor shall submit proof to the procuring activity that the saw complies to the requirements of the applicable UL standards. Compliance shall be indicated by the permanent attachment of an appropriate label, seal, or stamp that contains the UL listing mark to the product, or by a certificate furnished therewith. In lieu of the aforementioned, the contractor shall submit independent proof, satisfactory to the Government, that the saw complies to the requirements of the applicable UL standards including the UL method of test. Contractors compliance with the UL requirements does not absolve the contractor from complete compliance with

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the requirements of this specification. In the event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern. When electromagnetic interference (EMI) suppression is required, only the basic saw, prior to modification for EMI suppression, shall comply to the requirements of the applicable UL standard.

3.1.1 Design. The saw shall be either double-insulated or of the three-wire grounding construction.

3.2 First article inspection. When specified (see 6.1), the contractor shall furnish a completely fabricated portable electric saw for first article inspection in accordance with section 4. The sample shall be a pre-production item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used in production of items to fulfill the items of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.3 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of saws otherwise conforming to this specification.

3.4 Material. Material used in the construction of the saws shall be new and of good commercial quality suitable for the intended purpose. All materials shall be free from defects that may adversely affect the serviceability of the finished product.

3.5 Screw threads. Threaded parts including screws, bolts, and nuts shall conform to the applicable requirements of FED-STD-H28.

3.6 Lubrication. Bearings, gears, and moving parts shall be assembled with lubricating medium. If lubricant replenishment is required, external lubricant fittings or other suitable means shall be provided. Sealed-for-life bearings and permanently lubricated gear boxes shall not be required to have external fittings. The gears shall operate in a lubricant which shall be prevented from excessive leakage from the gear casing or entering the motor enclosure. Parts requiring protection from dust shall be enclosed.

3.7 Bearings. All rotating parts of the saw shall be provided with commercially available rolling contact bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of the lubricant and entrance of dirt and dust.

3.8 General construction. The saws shall withstand, without failure, treatment likely to be encountered under general service conditions. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. All parts shall be properly finished to dimensions within limits established by good commercial practice. Similar parts, of the same manufacture, shall be interchangeable without alteration with related parts having the same manufacturer's part number.

3.9 Housing. The housings shall be constructed to prevent failure under service conditions. Cap screws, if used, shall engage threads in ferrous housings, or ferrous inserts in nonferrous housings, by not less than a length equal to the diameter of the fastener thread. Where ferrous inserts are not used in non-ferrous housings, cap screws shall engage threads by not less than a length equal to 1-1/2 times the diameter of the fastener thread. All threaded fasteners shall be secured with locking devices or shall be installed with means of preventing loss of tightness.

3.10 Motor. The motor of the reciprocating saws shall be of the universal type designed to operate on either 120 volt (± 10%) direct current (DC), or 120 volt (± 6%) single-phase, 25-60 cycle alternating current (AC).

3.10.1 Commutator. The commutator segments shall be of hard drawn, hard rolled, or drop forged copper. The commutator shall have sheet mica separators between segments or shall be of the molded mica construction using material that will be immune to deformation and dielectric degradation when subjected to maximum rotational forces operating temperatures encountered in service.

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3.11 Reciprocating mechanism. Each saw shall be provided with a reciprocating mechanism to change the rotary motion of the motor to reciprocating motion for the blade.

3.12 Wiring. All wiring, flexible cord, attachment plug cap, and grounding means shall be in accordance with the requirements of UL 45. The cord shall have an attachment plug cap and, unless otherwise specified (see 6.1), a minimum free length of 8 feet measured from point of entry into the saw.

3.13 Handle. The handle of the reciprocating saws shall be of the closed grip type, similar to those illustrated. The handle shall be of such a size and shape as to provide a firm and comfortable grip. Type I saws shall be furnished with an additional handle for two-handed gripping. The type I saws may be furnished without the additional handle, provided the housing at the sawing end is small enough for firm hand gripping.

3.14 Switch. The saws shall be furnished with a conveniently located switch in the handle. The switch may have a lock-on mechanism provided that actuation and turn-off can be accomplished by the same finger(s) that actuate the switch.

3.15 Electromagnetic interference suppression. When specified (see 6.1), each saw shall be electromagnetic interference suppressed within the limits specified for portable electric tools in compliance with the requirements of MIL-STD-461.

3.16 Saw blade chuck or holder. The saw blade chuck or holder shall allow for easy blade changing without the use of special tools. The type I saw shall be designed to hold the type II blade conforming to GGG-B-425. The type II saw shall be designed to hold the type I blade conforming to GGG-B-425.

3.17 Carrying case. When specified (see 6.1), the saw shall be furnished in a durable sheet metal or molded plastic carrying case. The case shall be provided with a handle, hinged cover and at least one suitcase snap latch or equivalent means of holding the cover in place. The holder shall not open when the case and contents are carried by the handle. The case shall hold the saw and its accessories. Accessories shall consist of standard equipment (blades, wrench, wrench holder, etc.) as normally furnished commercially by the manufacturer.

3.18 Type I. Horizontal drive reciprocating saw.

3.18.1 Class 1. Each class 1 saw shall be similar to Figure 1 and meet the minimum requirements as specified herein and as shown in table I. The saw shall be of the variable speed design and shall be suitable for sawing either metal or wood. The no-load surface speed of the saw blade shall be not less than 1600 nor more than 2400 inches per minute, determined by multiplying the length of stroke in one direction by the number of strokes per minute. The saw blade shall reciprocate from the front end of the tool so that the blade and handle center are in general alignment, the same as any conventional hand saw. However, at the option of the contractor, the tool may provide for additional positioning of the blade. A saw guide or other means shall be provided whereby the tool can be guided.

3.18.2 Class 2. Each class 2 saw shall be in accordance with 3.18.1 except that the saw shall have at least two sawing speeds: "High" for fast sawing of wood and other soft materials, and "Low" for sawing either metal or wood. The speed selector shall be plainly marked "High" or "Low". The "High" no-load speed shall be not less than 2100 nor more than 2900 inches per minute. The "Low" no-load speed shall be at least 400 inches per minute slower than the "High" speed.

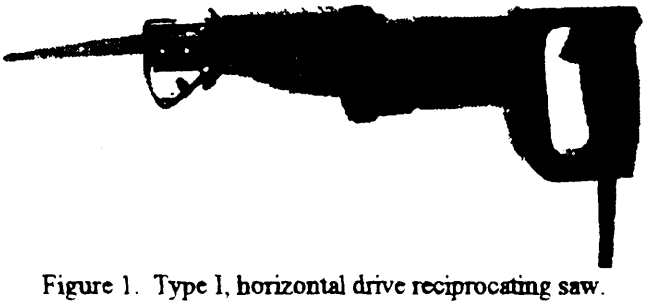


Figure 1. Type I, horizontal drive reciprocating saw.

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Table I. Portable electric reciprocating saw

Type	Minimum length of stroke (inches)	Maximum weight (pounds)	Maximum temperature rise after 10 successive duty cycles (°C)				Minimum average cutting rate through 3/16 inch steel strip 5 duty cycles (inches per minute)
			Motor Parts				
			Handles	Housing	Field Windings	Com-mutator	
I	7/16	9	20	50	65	75	3
II	7/16	7	20	50	65	65	2

3.19 Type II, vertical drive reciprocating saw. The type II saw shall be similar to figure 2 and shall meet the requirements specified herein and as shown in table I. The saw shall be of the variable speed design. The no-load surface speed of the saw blade shall be not less than 1600 nor more than 2400 inches per minute.

3.19.1 Base. Each type II saw shall be provided with a tilting base, adjustable for cutting angles up to 45°. Stops or markings shall be provided for angle settings at 90° (vertical), 30°, and 45°, both right and left.

3.19.2 Directional blower. Each type II saw shall be furnished with a blower to clear chips away from the cut.

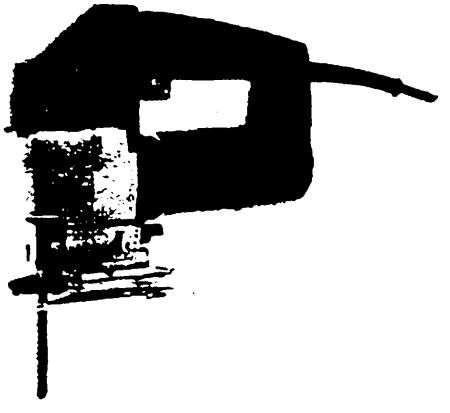


Figure 2. Vertical drive reciprocating saw.

3.20 Cutting capability. The type I and type II saws shall be capable of cutting under practical service conditions; hardwood with a minimum 1-3/4 inch thickness; aluminum alloy with a minimum 1/2 inch thickness; plastic, of the fabric base phenolic type, with a minimum 1/2 inch thickness; and mild steel with a minimum 3/16 inch thickness.

3.21 Accessories or extra equipment. When additional accessories or equipment are required, they shall be as specified (see 6.1).

3.22 Instruction book. Unless otherwise specified (see 6.1), one copy of an instruction book consisting of the manufacturer's standard commercial instructions and parts list shall be furnished with each saw.

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3.23 Identification plate. Each saw shall bear an identification plate in accordance with the applicable requirements of UL 45.

3.24 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.25 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for ensuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.2. Failure of the first article to pass the examination on one or more pre-production tests shall be cause for rejection. Unless otherwise specified (see 6.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All saws of the same type and size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample saw selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

(a) Inspection of the end item (see 4.5).
(b) Tests (see 4.6).
(c) Inspection of preparation for delivery requirements (see 4.7).

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4.5 Inspection of the end item.

4.5.1 Visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in terms of saws. the inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects.

Table II. Classification of defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Type and class not as specified	101	
	Lubricated parts enclosure nonconforming	102	
	Nuts, bolts or screws not secured by suitable locking device	103	
	Similar parts not dimensionally and functionally interchangeable	104	
	Nameplate nonconforming, illegible, incorrect, or missing	105	
	Handles not as specified, missing, or damaged	106	
	Switch and locking device not as specified	107	
	Motor not as specified	108	
	Cable and attachment plug nonconforming	109	
	Carrying case (when specified) nonconforming	110	
	Base (type II) not as specified	111	
	Directional blower (type II) not as specified	112	
	Saw blade chuck or holder not as specified	113	
	Accessories not as specified	114	

4.5.2 Dimensional examination. The end items shall be examined for any nonconformance to the dimensional requirements specified. The lot size shall be expressed in terms of saws. The sample unit shall be one saw. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.5.3 Tests for quality conformance. Each sample selected in accordance with 4.3.2 shall be tested for compliance with the requirements of this specification. The AQL shall be 2.5 percent defective. Nonconforming saws shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the sample tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.6.

4.6 Tests.

4.6.1 Electromagnetic interference test certification. If an EMI suppression network is required, the contractor shall provide certification that the suppression network in the saws offered conform to the requirements of 3.15 or are identical to that of saws previously tested and approved.

4.6.2 UL 45 certification. The offerer shall provide certification that the saw offered conforms to the test requirements of UL 45 for this particular tool (see 3.1 for compliance to this test).

4.6.3 Speed, temperature, and other tests. Sample saws shall be tested to verify that the saws comply with the requirements specified below and as listed on the nameplate. Variation in values shall not be more than percentages allowed by UL 45. Nonconforming saws shall be rejected:

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(a) No-load blade speed.
(b) Minimum length of stroke.
(c) Maximum temperature rise per Table I requirements.
(d) Minimum cutting rate per Table I requirements.

4.6.4 Cutting test. One saw shall satisfactorily complete one hour of actual work cutting materials of the type and sizes specified in 3.20 using saw blades as recommended by the manufacturer for the material being cut. The cuts shall be at least two inches long, cutting first the wood, then aluminum, then plastic, and then the steel. This varying cycle (at 15 minute intervals) shall continue until the one hour duration is complete. Upon completion of the cutting test, the saw shall be disassembled and the various parts examined. Broken, cracked, or deformed parts shall be cause for rejection.

4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Table III. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, each unit package and shipping container shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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a peak-to-peak double amplitude from 5.5 to 7.0 Hz

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style, if applicable, of item required (see 1.2).
c. First article approval, if required (see 3.2 and 4.2).
d. Cord length required, if different (see 3.12).
e. If electromagnetic interference suppression is required (see 3.15).
f. If carrying case is required (see 3.17).
g. Accessories or extra equipment required (see 3.21).
h. Number and details of instruction book(s) required (see 3.22).
i. Level of preservation, packaging, and packing required (see 5.1).
j. Marking, if other than specified.

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